

Human Body Systems: Breathing & Transport

Answer sheet · 38 marks total

Section A — Respiration

A1. Carbon dioxide (1) + water (1).

A2. Any two: aerobic needs oxygen, anaerobic does not (1); aerobic releases (much) more energy per glucose molecule (1); anaerobic produces lactic acid, aerobic produces carbon dioxide and water (1).

A3. Anaerobic respiration (1); it produces lactic acid, which builds up and causes muscle fatigue/cramp (1).

Section B — Exchange surfaces

B1. Any three: large surface area (1); thin walls/short diffusion distance (1); good blood supply (1); moist surface/lining (1).

B2. Millions of alveoli give a large surface area (1); walls are one cell thick, giving a short diffusion distance (1); surrounded by a network of capillaries, giving a good blood supply that maintains a steep concentration gradient (1); a moist lining lets gases dissolve (1).

B3. This gives a short diffusion distance/pathway (1), so substances can diffuse quickly between the blood and body cells (1).

Section C — The heart & circulation

C1. Right atrium (1), right ventricle (1), left atrium (1), left ventricle (1).

C2. Blood passes through the heart twice on each full circuit of the body (1): once to the lungs (pulmonary circuit) and once to the rest of the body (systemic circuit) (1).

C3. The left ventricle pumps blood all the way around the body (systemic circuit) (1), while the right ventricle only pumps blood to the nearby lungs, so the left ventricle needs to generate much higher pressure (1).

Section D — Blood vessels

D1. E.g. arteries have thicker, more muscular/elastic walls (1) because they carry blood at high pressure away from the heart (1). [*or: veins have a wider lumen (1) to help blood flow at low pressure (1)*]

D2. Blood in veins is at low pressure (1); valves stop the blood from flowing backwards (1).

Section E — Blood components

E1. Red blood cells (1), white blood cells (1), platelets (1), plasma (1).

E2. Biconcave disc shape gives a large surface area (1); no nucleus, giving more space to carry oxygen/haemoglobin (1); contains haemoglobin, which binds to oxygen (1).

E3. They help the blood clot at a wound (1).

Section F — Apply it

F1. Fewer red blood cells means less haemoglobin to carry oxygen (1); so less oxygen is transported to (respiring) body cells (1); less oxygen means less energy released by (aerobic) respiration (1).

Marking note: accept any scientifically correct alternative answer with the same reasoning, even if different from the example shown — award marks for a correctly linked point, not for matching these exact words.